

Project Proposal Preview

Request: I want to add a search bar in my website

1. Requirements

Add a search bar to the website allowing users to query content and view results.

Functional requirements

- Provide an input field with placeholder text.
- Include a submit button or allow Enter key to trigger search.
- Send query to a search API endpoint via AJAX.
- Display results in a list below the search bar.
- Show a loading indicator while awaiting response.
- Handle no-results case with a user-friendly message.

Non-functional requirements

- Responsive design for mobile and desktop.
- Accessible: keyboard navigation and ARIA labels.
- Fast response: results load within 2 seconds.
- Compatible with latest Chrome, Firefox, Safari, Edge.
- Minimal impact on page load time.
- Maintainable code with clear separation of concerns.

Edge cases

- Empty query triggers validation error.
- Query with special characters handled safely.
- Very long queries truncated or limited.
- Network failure shows error message.
- Rapid typing does not overwhelm API (debounce).
- Results exceeding viewport scroll appropriately.

Assumptions

- A search API endpoint exists and returns JSON.
- User authentication is not required for search.
- Search results are simple text snippets.
- No pagination needed for initial implementation.
- Browser supports ES6 features.
- No server-side rendering required.

2. Architecture

Add a vanillaJS search bar that debounces input, calls the existing search API via fetch, and renders results dynamically.

Component	Responsibility	Type
SearchBar	Render input field, handle user input, validation, debounce, trigger search request	frontend

Component	Responsibility	Type
SearchResults	Display loading state, list of results, no result or error messages	frontend
SearchService	Encapsulate AJAX call to the search API and return parsed JSON	integration

Data flow

- User focuses the input field
- User types query – debounce prevents rapid calls
- On Enter or button click, SearchService sends fetch request to API endpoint
- Loading indicator shown by SearchResults
- API responds with JSON or error
- SearchResults parses response and renders list or appropriate message
- If network error, display error message

File structure

- index.html
- css/search.css
- js/search.js

3. Tools

Tool	Category	Purpose	Required
lodash.debounce	frontend	Provide debounce function to limit API calls during rapid typing	Yes

4. Cost Estimate (tools/services only)

Tool	Pricing model	Est. cost	Notes
lodash.debounce	free	\$0	Open-source library, no cost

Estimated monthly total: \$0

Note: These are third-party tool/service cost estimates only, not a cost of building the feature